Facial Recognition Attendance System (FRAS): Midterm Design Report

Ricardo J. Lopez Ortiz and Miguel A. de Jesús Rosa
Mentor: Prof. Alcides Alvear
CPEN 491 – Senior Design Project I · Engineering, Design and Architecture · Universidad Ana G. Méndez, Gurabo, Puerto Rico

***Abstract*—Taking attendance by calling roll or passing a sign-in sheet consumes class time and can produce errors in the attendance record. This report presents the midterm progress in the design of the Facial Recognition Attendance System (FRAS), which records class attendance by recognizing, in images captured by a classroom camera, the faces of the students enrolled in a section. The professor starts and closes each session and reviews the list through a web page, and an administrator enrolls the facial data of the students and maintains the course data obtained from the university information system. The report describes the literature search, the project management approach, a specification of 128 requirements with measurable targets of at least 90% recognition accuracy and less than 3 s response time, a technology-independent structural diagram of seven subsystems, twelve use cases traced to the functional requirements, and ten wireframes of the web interface. The hardware and software have not been selected: a study of alternatives defines the options for image capture, processing location, recognition software, data storage, and connection with the university, together with the criteria to select them, with cost as the main criterion.**

***Index Terms*—Attendance system, biometrics, face recognition, privacy, requirements engineering, use cases.**

I. Introduction

In many university courses, attendance is taken by calling roll or by passing a sign-in sheet. Both methods use up class time, and both can lead to errors in the attendance record: a name may be skipped or misheard, a student may sign for a classmate who is absent, and paper sheets must later be transcribed by hand.

In Presentation #1 the team compared two solutions to this problem: a dynamic QR code generated by each student in the Blackboard application and scanned on arrival, and a facial recognition attendance system. The second solution was selected because it requires no action from the student. Its operation has four stages. A camera installed at the front of the classroom captures images of the students during class (Fig. 1); the system detects the faces in each image and compares them with the faces of the students enrolled in the section; it records the attendance with the date and time; and the professor reviews and, if necessary, corrects the list through a graphical interface. The professor never interacts with the camera directly.

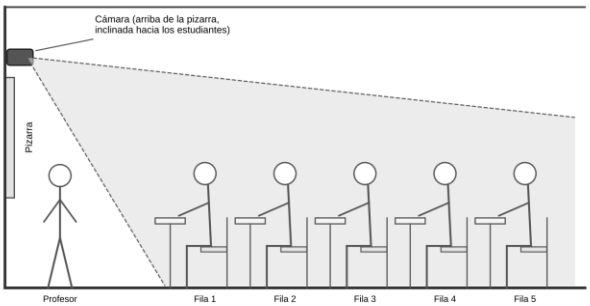


Fig. 1. Illustrative position of the classroom camera (side view).

A system of this kind handles biometric data of students, which are protected by federal guidance in the United States [1] and by international standards [2]. For this reason, privacy is treated as a requirement and as a selection criterion from the beginning of the design. In addition, the team must keep the cost of the system low, and the hardware and software that will implement it have not yet been selected; the design presented here is therefore independent of any specific technology.

The rest of the report is organized as follows. Section II describes the literature search and related work. Section III presents the project management approach. Section IV summarizes the requirements specification. Section V presents the structural diagram, Section VI the use cases, and Section VII the wireframes of the user interface. Section VIII describes the study of alternatives for the hardware and software. Section IX lists the next steps, and Section X concludes the report.

II. Literature Search and Related Work

A. Search Method

The search followed the methodology presented in the course: sources were located in digital indexes, the queries were recorded, and the quality of each source was judged by its venue and its number of citations. The search was performed in IEEE Xplore and Google Scholar, for publications from 2021 to 2026, with earlier foundational works accepted. A source was included only if it was peer reviewed or normative, if its full text could be read by the team, and if it supported a requirement or a design decision. Retracted articles and works on masked faces or COVID-19 conditions were excluded. Table I records the main queries.

TABLE I
Search Log

Index	Query	Filters	Results	Selected
IEEE Xplore	("face recognition" AND attendance)	2021–2026	889	—
IEEE Xplore	("face recognition" AND attendance)	2021–2026, journals	11	[3], [4]

Index	Query	Filters	Results	Selected
IEEE Xplore	("face detection" AND "Raspberry Pi")	2021–2026, journals	21	[5], [6]
IEEE Xplore	("face recognition" AND attendance AND "Raspberry Pi")	2021–2026, journals	0	—
IEEE Xplore	(biometric AND privacy AND (education OR student))	2021–2026, journals	107	[7]
Google Scholar	"face recognition" attendance classroom	Since 2021	First page	[8]
Google Scholar	Title searches (survey, YuNet, NIST IR 8280)	Title search	1 each	[9], [10], [11]

Most results are conference papers; only 11 journal articles on face-recognition attendance appear in IEEE Xplore since 2021, and one of them is a retraction notice, which was excluded. Three sources considered earlier were replaced because their full text was not available to the team.

B. Related Work

Automatic attendance has been approached in several ways. Gao *et al.* [4] proposed a crowdsensing method in which the server asks students to submit their location and then asks selected students to count classmates in sub-regions of the classroom; it avoids biometrics but still requires action from the students. Face-recognition approaches remove that burden. Raj *et al.* [3] built an attendance system in which a camera captures students as they enter, detects faces with the HOG algorithm, encodes each face as 128 measurements, stores the records in an SQLite database, and e-mails the attendance to an administrator. Ismail *et al.* [8] developed a web-based university classroom attendance system with open-source pre-trained deep-learning models that stores the extracted feature vectors in an online database; in user tests with frontal and 30–45° side views it reported 92% accuracy, 100% precision, and 90% recall. These works were used to check the functional scope of the FRAS requirements, and they support storing feature vectors instead of images.

Rukhiran *et al.* [5] built IoT-based biometric systems to verify the identity of hundreds of undergraduate exam takers, using a single-board computer with a camera connected to a database server, and evaluated them by accuracy rate, error rate, response time, and cost. They found that a unimodal facial system gives excellent support for identity verification, which supports the use of face recognition alone in FRAS, and their evaluation metrics are reused in the selection criteria of Section VIII.

Modern recognizers map each face to a fixed-length vector, or embedding, so that faces of the same person lie close together and identity can be decided with a distance threshold [12]; Wang and Deng [9] survey the architectures, loss functions, and datasets behind this approach. Baobaid *et al.* [6] review the hardware used for real-time face recognition and compare algorithms by accuracy and processing time. Lightweight models such as the YuNet detector [10] and the SFace recognizer [13], distributed with OpenCV [14], are among the candidates evaluated in the study of alternatives.

Face recognition also carries privacy risks. Meden *et al.* [7] review privacy-enhancing techniques for face biometrics that address legal requirements. FERPA guidance treats facial characteristics as a protected biometric record [1], and ISO/IEC 24745 [2] requires confidentiality and revocability of stored biometric data. NIST IR 8280 [11] documents differences in error rates across demographic groups, which motivates a requirement to monitor recognition errors by group. These sources make privacy a criterion for every design alternative, in particular for where images are processed and stored.

III. Project Management

The project is developed by a two-member team, Ricardo J. Lopez Ortiz and Miguel A. de Jesús Rosa, who share all tasks of analysis, design, and documentation, under the mentorship of Prof. Alcides Alvear. The team follows the evolutionary prototyping life-cycle model [15], [16]: non-functional prototypes—GUI wireframes, diagrams, and use-case walkthroughs—are reviewed with the mentor and with professors, and their feedback is used to revise the requirements and the design. This model was chosen because the acceptance of an attendance system depends on the preferences of many professors, which are easier to obtain and reconcile when they react to a concrete prototype. Table II lists the milestones of the semester.

TABLE II
Milestones

Date	Milestone	Status
—	Problem definition and Presentation #1	Complete
—	Requirements specification and literature search	Complete
Oct. 20, 2026	Midterm report	Complete
Oct. 22, 2026	Informal review with the professor	Planned
Oct. 27, 2026	Presentation #2	Planned
Oct.–Nov. 2026	Interviews with professors; update of requirements and use cases	Planned

IV. Requirements

A. How the Requirements Were Obtained

The requirements were derived in three steps. First, the functional requirements (FR) were written from the problem definition and the proposed solution—capture, recognize, record, and review—and grouped by feature area. Second, the non-functional requirements (NFR) were organized under the nine product-quality characteristics of ISO/IEC 25010:2023 [17], and the document follows the structure of IEEE Std 830 [18]. Third, the security and safety requirements were grounded in regulatory guidance and research on biometric data [1], [2], [11], [7]. The functional requirements were then compared with the systems described in Section II to confirm that no essential function was missing, and the review of the wireframes (Section VII) added requirements for the preview and deletion of captured images, the start of a session only for the class scheduled at that time, the import of course data, the management of user accounts, and the entry of authorized students from other sections.

B. Summary

The specification contains 128 requirements: 57 functional and 71 non-functional. Requirements written with "shall" are mandatory (40 FR, 29 NFR); those written with "should" are desirable and form the priority list for trade-offs (17 FR, 42 NFR). Table III gives the count per group with one example of each.

The complete list of requirements, with their priorities, measurable targets, and traceability to the use cases, is published in the FRAS Software Requirements Specification [19].

TABLE III
REQUIREMENTS PER GROUP

Group	Total	Example
FR – Face detection and recognition	10	FR-03: The system shall recognize registered individuals using stored facial data.
FR – User registration and management	9	FR-12: The system shall capture facial data during user registration.
FR – Attendance management	15	FR-20: The system shall automatically mark attendance after successfully recognizing a registered person.
FR – Records and reports	7	FR-37: The system should provide the ability to export attendance reports to formats such as Excel or PDF.
FR – Attendance sessions	5	FR-56: A professor shall be able to start a session only for the section scheduled at that time, and only one session at a time.
FR – Camera operations	2	FR-46: The system shall indicate whether the camera is connected and operational.
FR – Authentication and access	4	FR-47: The system shall provide an authentication mechanism for administrators.
FR – Database and system operations	5	FR-50: The system should provide a mechanism for backing up attendance and registered-user information.
NFR – Functional suitability	5	NFR-05: Attendance records shall correctly correspond to the individuals recognized by the system.
NFR – Performance efficiency	6	NFR-07: The system should identify a registered face within a predefined response time.
NFR – Compatibility	4	NFR-12: The application shall operate with the selected camera hardware.
NFR – Interaction capability	8	NFR-19: The interface shall clearly communicate whether recognition and attendance recording were successful.
NFR – Reliability	18	NFR-29: Repeated facial scans shall not generate unintended duplicate attendance records.
NFR – Security	13	NFR-46: Facial images, embeddings, or templates shall be protected from unauthorized access.
NFR – Maintainability	6	NFR-59: The recognition algorithm should be replaceable or upgradeable without complete system redesign.
NFR – Flexibility	7	NFR-61: The system should maintain acceptable performance as the number of registered users increases.
NFR – Safety	4	NFR-69: The system shall provide a documented manual attendance procedure so that no student is unfairly marked absent solely due to a recognition-system failure.
Total	**128**	

C. Proposed Measurable Targets

Several non-functional requirements use words such as "acceptable" or "reasonable," which cannot be tested. Table IV proposes measurable targets for them. These are design targets to be confirmed with the professors and verified during testing; they are not measured results. The accuracy target is based on the 92% reported by Ismail *et al.* [8].

TABLE IV
PROPOSED MEASURABLE TARGETS

ID	Requirement	Proposed target	Reference point
NFR-01	Face recognition accuracy	≥ 90% of present, enrolled students correctly identified per session	Ismail *et al.* report 92%
NFR-03	False acceptance rate	≤ 1% of automatically confirmed records belong to the wrong student	Team proposal
NFR-06	Face detection speed	≤ 0.5 s per captured frame on the selected hardware	Team proposal
NFR-07	Face recognition speed	< 3 s from capture to stored record	Team target
NFR-09	GUI response time	≤ 1 s for navigation actions	Team proposal
NFR-10	Database query performance	≤ 2 s to search one semester of records	Team proposal
NFR-11	Application startup time	≤ 60 s from power-on until the system is ready	Team proposal
NFR-21	Minimal training	A professor completes a session (start, review, close) after a 10-minute demonstration	Team proposal
NFR-31	Distance robustness	Recognition up to the last row of the pilot classroom	To be set after measuring the classroom
NFR-61	User scalability	No increase in recognition time for sections of up to 60 students	Team proposal

Suggestions from the mentor about how a session should start (automatically at the class time, manually, or both) and about

captures requested by the professor will be validated in structured interviews with professors before they are added to the specification.

V. Structural Diagram

Fig. 2 shows the structural diagram of FRAS, which represents the subsystems involved in the operation of the system and the order in which they work. It formalizes the system flow proposed in Presentation #1. After a valid login, the session-management subsystem opens an attendance session for a course section. The image-capture subsystem then takes images of the classroom, the face detection and recognition subsystem identifies the registered students in each image, and the attendance-recording subsystem stores the result with its date and time. Capture, recognition, and recording repeat until the session is over. The professor then reviews and corrects the attendance list, and the reports subsystem makes the records available for consultation and export. Table V describes each subsystem. The diagram is independent of the hardware and software that will implement each subsystem.

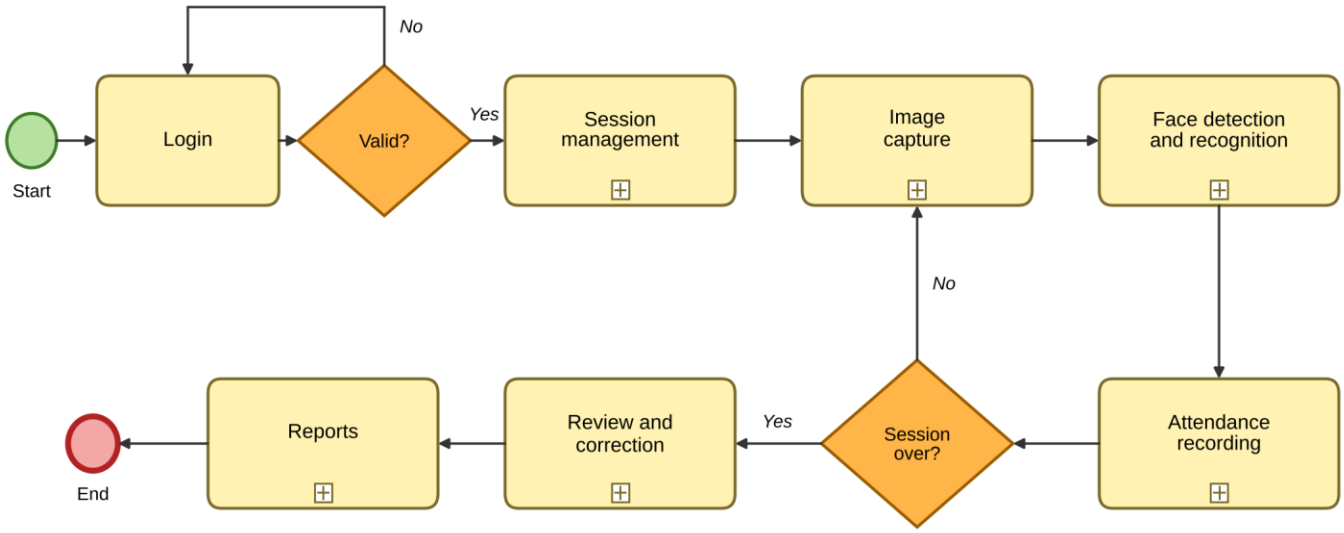


Fig. 2. Structural diagram of FRAS (subsystems marked ⊞ contain internal steps).

TABLE V
Subsystems of FRAS

Subsystem	Function
Login	Verifies the identity of the user (administrator or professor) and grants the functions allowed for that role.
Session management	Opens and closes an attendance session for a course section and requests image captures during the session.
Image capture	Takes images of the classroom when the session requests them.
Face detection and recognition	Finds the faces in each image and identifies the registered students of the section; faces that cannot be identified with enough confidence are flagged.
Attendance recording	Records each identified student as present or late with the date and time, without duplicates.
Review and correction	Lets the professor confirm the list, resolve flagged faces, and correct or enter attendance manually.
Reports	Presents attendance history and percentages and exports the records.

VI. Use Cases

Use cases describe the behavior of the system from the point of view of its users and define the boundary between the system and its environment [20]. The FRAS use-case model has two actors. The professor starts and closes attendance sessions, reviews and corrects the attendance list, and consults reports. The administrator enrolls the facial data of the students, synchronizes course and roster data with the university information system through a direct connection, and maintains user accounts and backups. FRAS does not create courses or students: they are managed by the university system and only read by FRAS. Students are not actors: they do not interact with the system, and their attendance is recorded automatically while a session is open. Logging in (UC-01) is a precondition of every use case. Fig. 3 shows the use-case diagram.

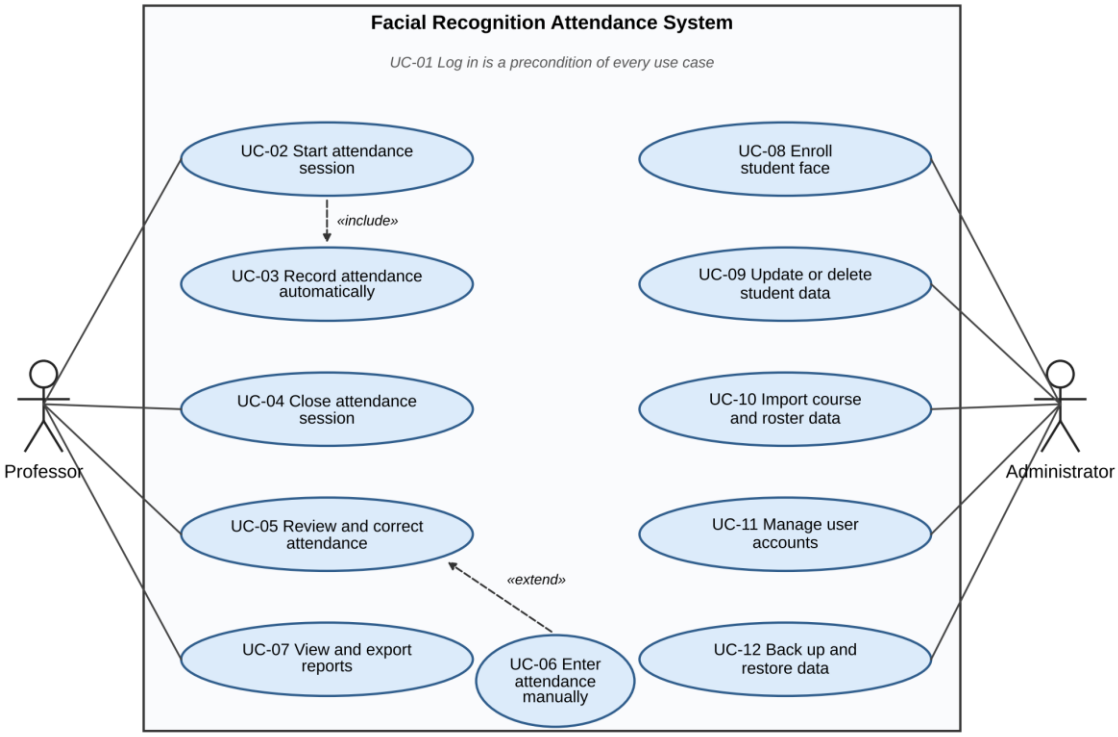


Fig. 3. Use-case diagram of FRAS.

A. User Stories

Each use case was first written as a user story in the form "As a ⟨role⟩, I want to ⟨goal⟩ so that ⟨benefit⟩" [21]:

1) As a professor, I want to start an attendance session for my section so that attendance is taken without calling roll. (UC-02, UC-03)
2) As a professor, I want to close the attendance session so that students who were not recognized are marked absent. (UC-04)
3) As a professor, I want to review and correct the attendance list so that the final record is accurate. (UC-05)
4) As a professor, I want to add a student from another section who was authorized to attend so that the student's attendance is also recorded. (UC-06)
5) As a professor, I want to view and export attendance reports so that I can use them in my course records. (UC-07)
6) As an administrator, I want to enroll a student's face so that the system can recognize the student in class. (UC-08)
7) As an administrator, I want to update or delete a student's data so that records stay correct and facial data can be removed when required. (UC-09)
8) As an administrator, I want to synchronize the course sections, schedules, and class rosters with the university system so that each session belongs to the right course, professor, and students. (UC-10)
9) As an administrator, I want to manage user accounts and back up the data so that each person has the correct access and records are not lost. (UC-11, UC-12)

B. Use-Case Descriptions

Table VI summarizes the twelve use cases, and Table VII describes UC-02, the core of the system, in detail.

TABLE VI
SUMMARY OF USE CASES

ID	Use case	Actor	Description
UC-01	Log in	Professor, Administrator	The user enters credentials and receives the functions allowed for the user's role.
UC-02	Start attendance session	Professor	The professor selects the section scheduled at that time and opens an attendance session; only one session can be open at a time. Whether a session can also start automatically will be decided after the interviews with professors.
UC-03	Record attendance automatically	System (included by UC-02)	While the session is open, the system captures images, recognizes the registered students of the section, and records them as present or late without duplicates. Doubtful matches are flagged for review.
UC-04	Close attendance session	Professor	The professor ends the session; students who were never recognized are marked absent.
UC-05	Review and correct attendance	Professor	The professor checks the list, resolves flagged matches, and corrects any record. When the professor confirms the attendance, the captured images are deleted.

ID	Use case	Actor	Description
UC-06	Enter attendance manually	Professor (extends UC-05)	The professor adds to the session a student from another section who was authorized to attend that class.
UC-07	View and export reports	Professor	The professor consults attendance history and percentages and exports them.
UC-08	Enroll student face	Administrator	For a student already present in the imported roster, the administrator informs the student about the use of facial data and captures reference images of the face.
UC-09	Update or delete student data	Administrator	The administrator modifies a student's data, updates the facial data, or deletes it.
UC-10	Import course and roster data	Administrator	FRAS obtains the sections, schedules, professors, and class rosters through a direct connection to the university information system; the administrator starts or schedules the synchronization.
UC-11	Manage user accounts	Administrator	The administrator creates, edits, and deactivates accounts and assigns their roles.
UC-12	Back up and restore data	Administrator	The administrator creates a backup of the data or restores it from a valid backup.

TABLE VII
DESCRIPTION OF UC-02 START ATTENDANCE SESSION

Field	Description
Actor	Professor
Precondition	The professor is logged in (UC-01); the section and its roster have been imported (UC-10) and its students have enrolled facial data (UC-08).
Main flow	1. The professor selects the section scheduled at that time. 2. The professor selects "Start session." 3. The system opens the session and records its start time. 4. «include» UC-03: the system captures images, recognizes the students, and records their attendance. 5. The system shows the professor the students recognized so far.
Alternate flows	A1. A session for that section is already open: the system shows the open session instead of creating a new one. A2. Image capture is not available: the system informs the professor, who records the attendance in the review (UC-05). A3. A face is recognized with low confidence: it is flagged for the professor instead of being recorded.
Postcondition	An attendance session is open and attendance is being recorded for the section.
Requirements	FR-40, FR-41, FR-56; through UC-03: FR-01–FR-10, FR-20–FR-27, FR-54

C. Traceability

Table VIII links each use case to the functional requirements of the specification. Every use case is covered by at least one functional requirement.

TABLE VIII
USE CASES AND FUNCTIONAL REQUIREMENTS

Use case	Functional requirements
UC-01 Log in	FR-47, FR-48, FR-49
UC-02 Start session	FR-40, FR-41, FR-43, FR-56
UC-03 Record automatically	FR-01–FR-10, FR-20–FR-23, FR-25–FR-27, FR-32, FR-44, FR-46, FR-54
UC-04 Close session	FR-24, FR-28, FR-42
UC-05 Review and correct	FR-29–FR-31, FR-33–FR-35, FR-52, FR-55
UC-06 Enter manually	FR-59
UC-07 Reports	FR-36, FR-37, FR-38
UC-08 Enroll face	FR-11–FR-15
UC-09 Update or delete	FR-16–FR-19
UC-10 Import course data	FR-57
UC-11 User accounts	FR-49, FR-58
UC-12 Backups	FR-50, FR-51, FR-53

VII. WIREFRAMES

The user interface of FRAS will be a web page used by two roles. Low-fidelity wireframes were drawn for each screen to validate the use cases before any technology is selected; they are in grayscale on purpose, since they show content, controls, and navigation, not the final visual design. The professor works with five screens (Figs. 4–8). After logging in, the professor sees his or her sections and can start a session only for the class scheduled at that moment. During the session, the professor sees the students recognized so far; there is no live camera view, but each automatically recorded student has a Preview button that shows the image captured of that student. Captured images are temporary and are deleted when the professor confirms the attendance. In the review screen, the professor corrects any status, confirms or rejects the faces flagged with low confidence, and adds by manual entry a student from another section who was authorized to attend. Reports can be filtered by section and dates and exported to Excel or PDF. The administrator works with five additional screens (Figs. 9–13): a dashboard, the facial data of the students, the course data read from the university system, the user accounts, and the backups. Table IX relates each screen to its use cases. The wireframes are also used in the interviews with professors.

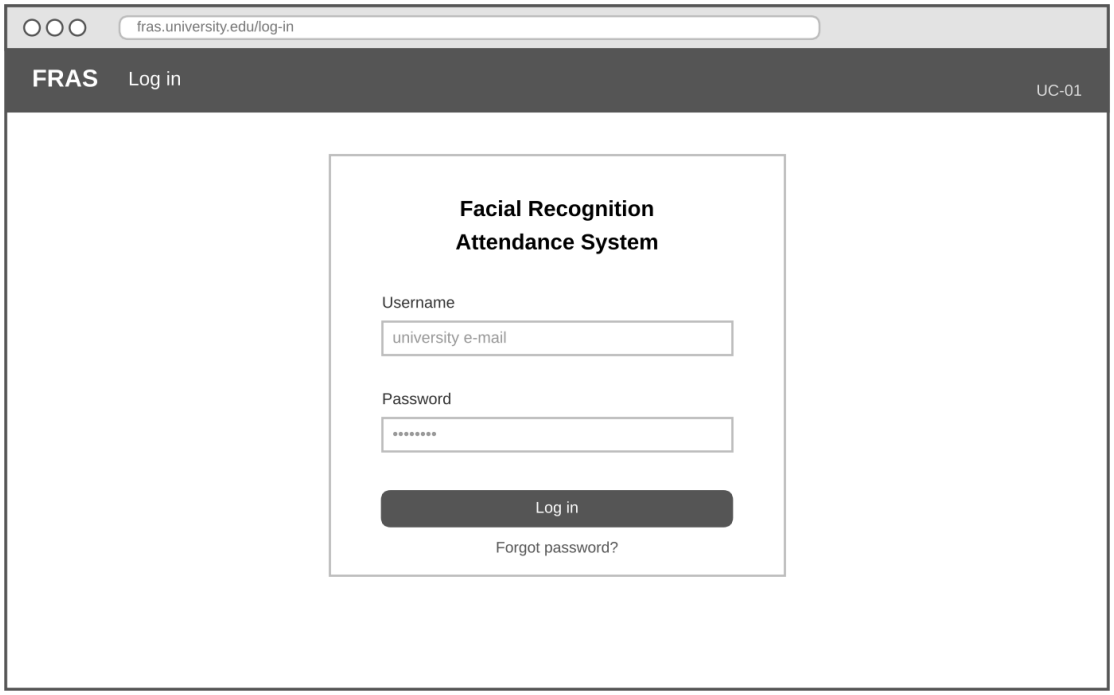


Fig. 4. Log-in screen, shared by both roles (UC-01).

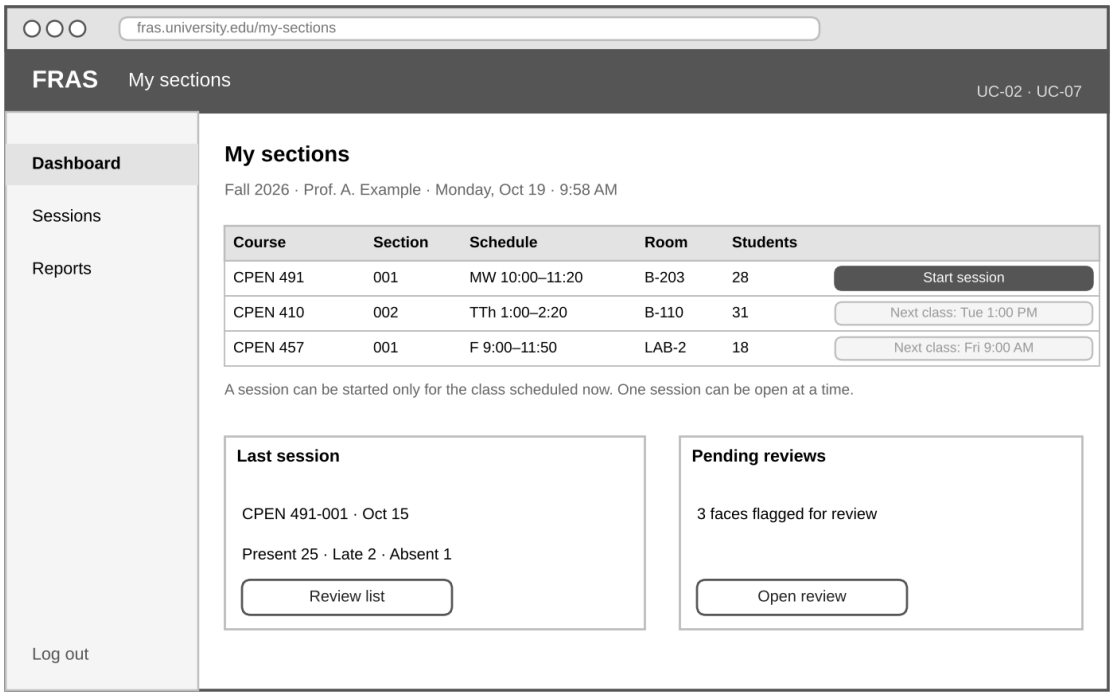


Fig. 5. My sections: only the class scheduled now can be started (UC-02, UC-07).

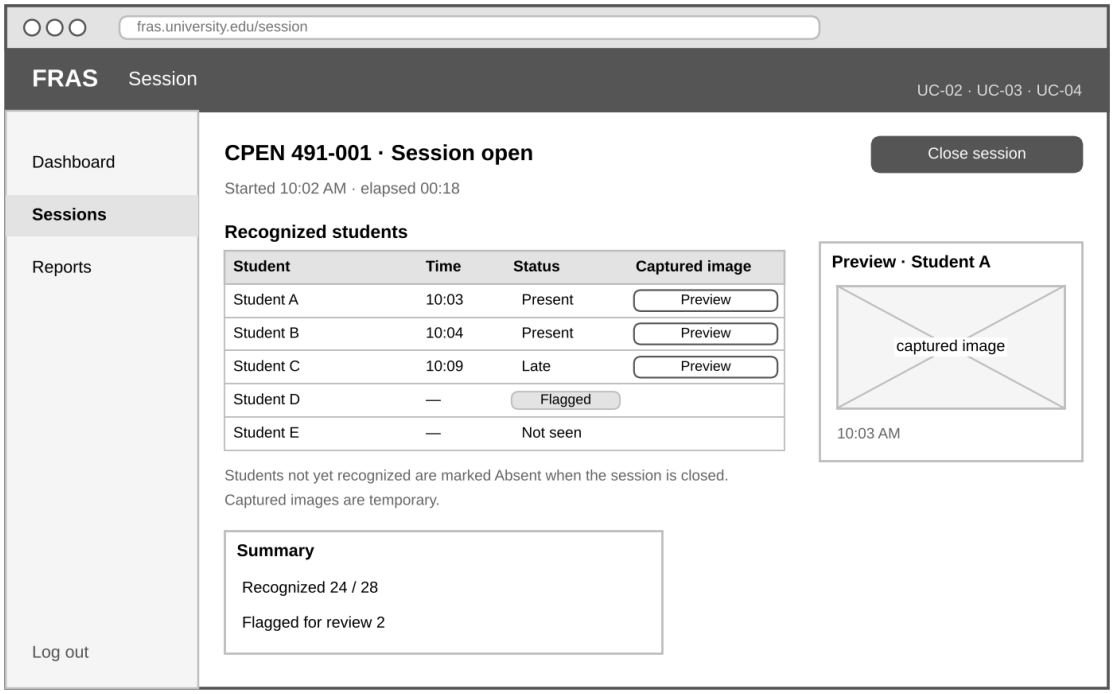


Fig. 6. Open session: recognized students and preview of the captured image (UC-02, UC-03, UC-04).

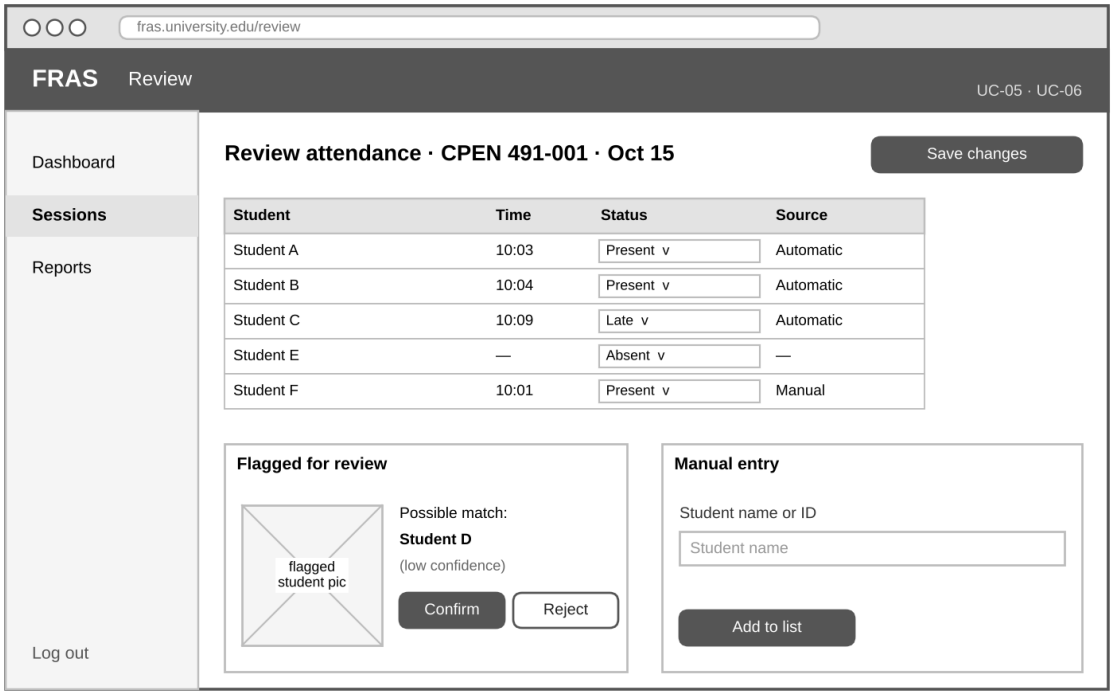


Fig. 7. Attendance review: status correction, flagged faces, and manual entry (UC-05, UC-06).

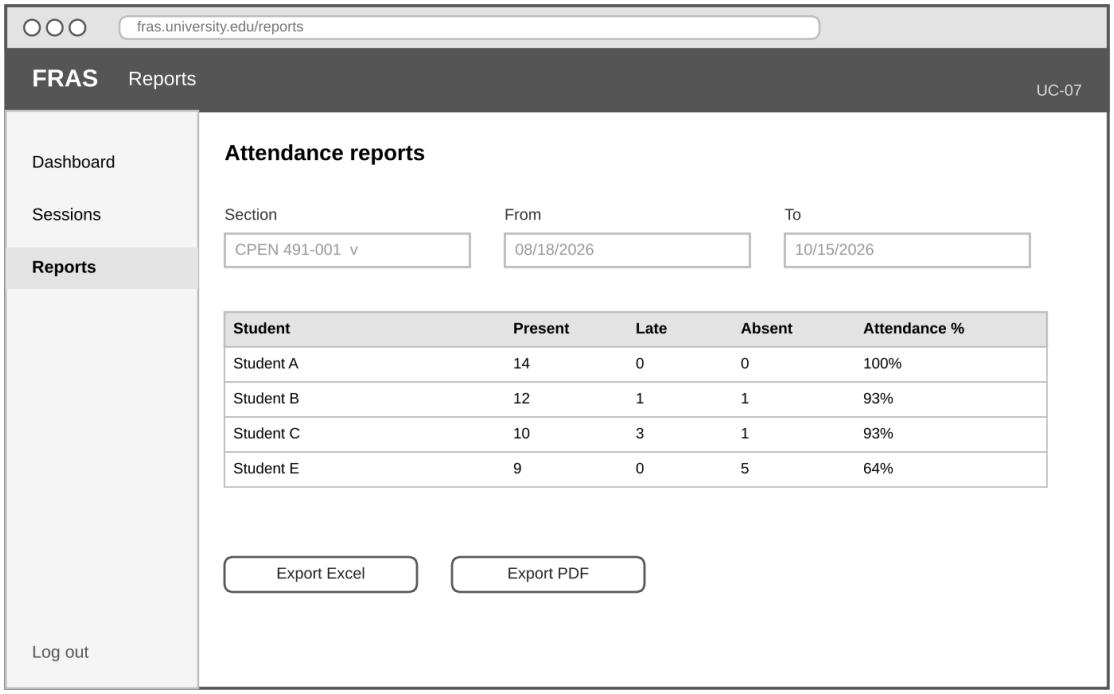


Fig. 8. Attendance reports and export (UC-07).

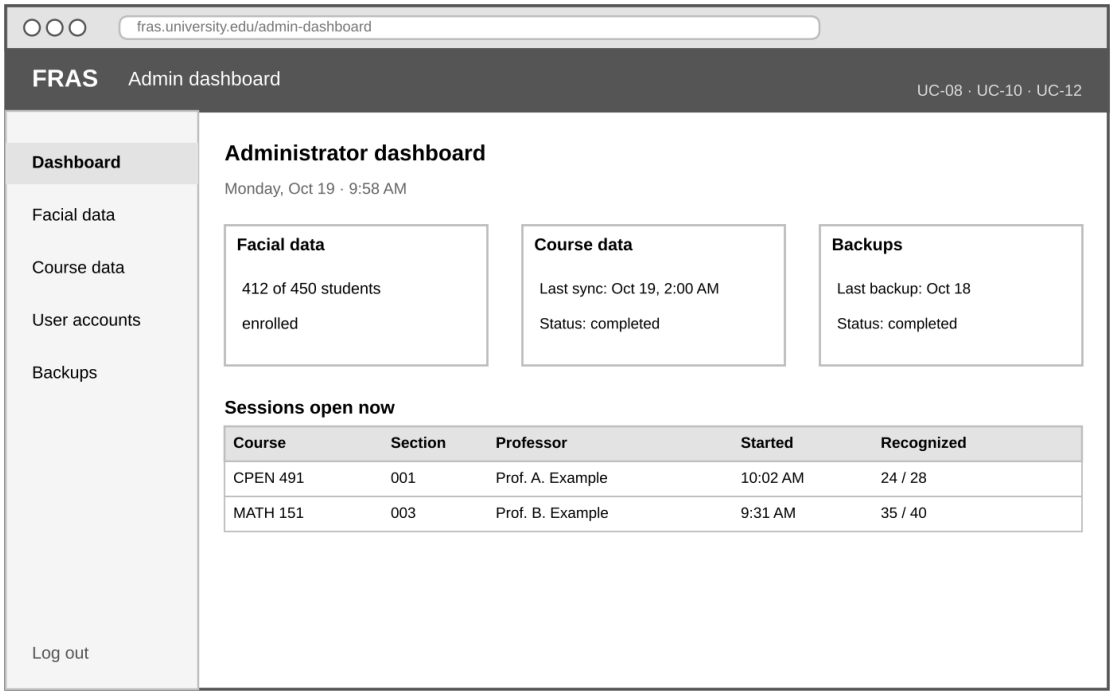


Fig. 9. Administrator dashboard (UC-08, UC-10, UC-12).

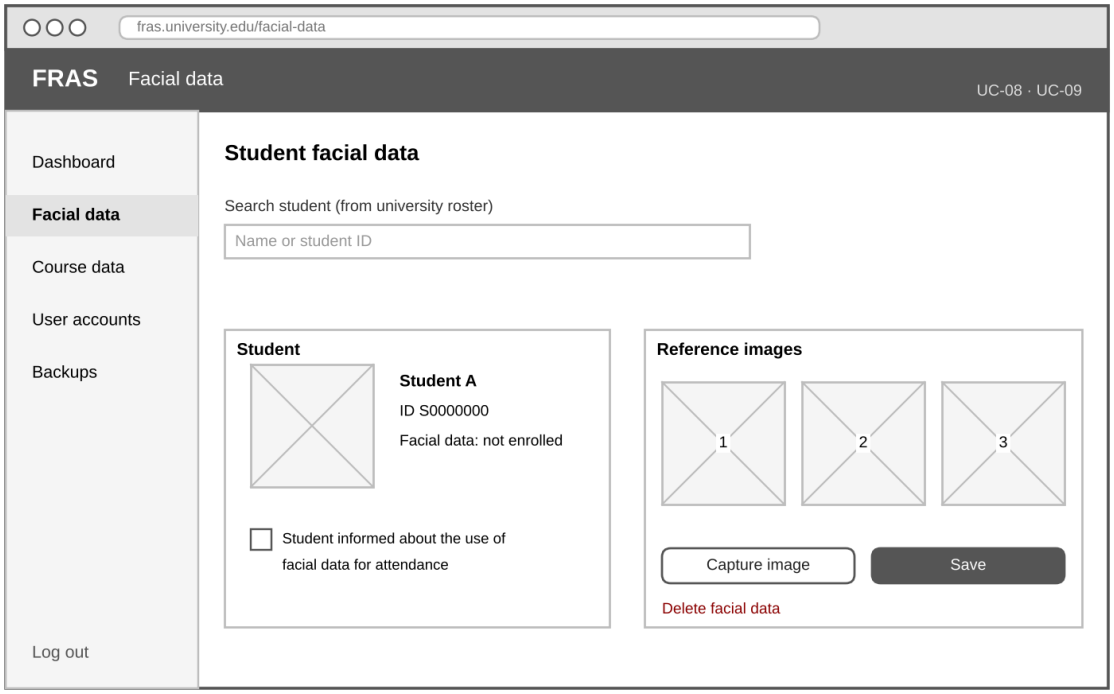


Fig. 10. Student facial data: enrollment, update, and deletion (UC-08, UC-09).

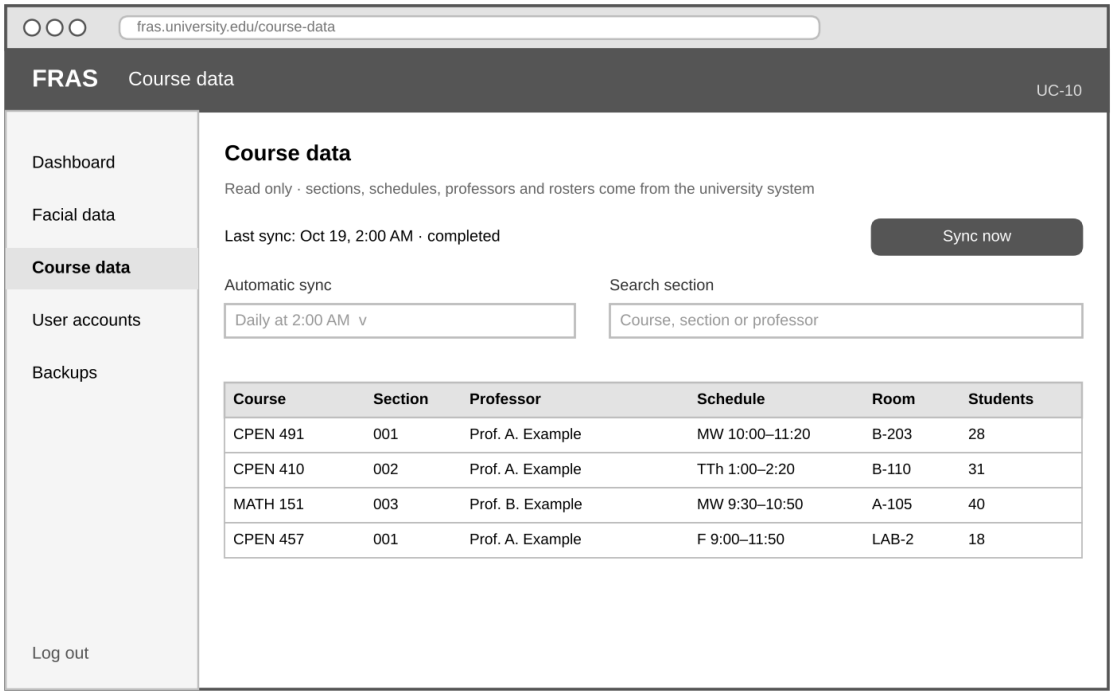


Fig. 11. Course data read from the university system (UC-10).

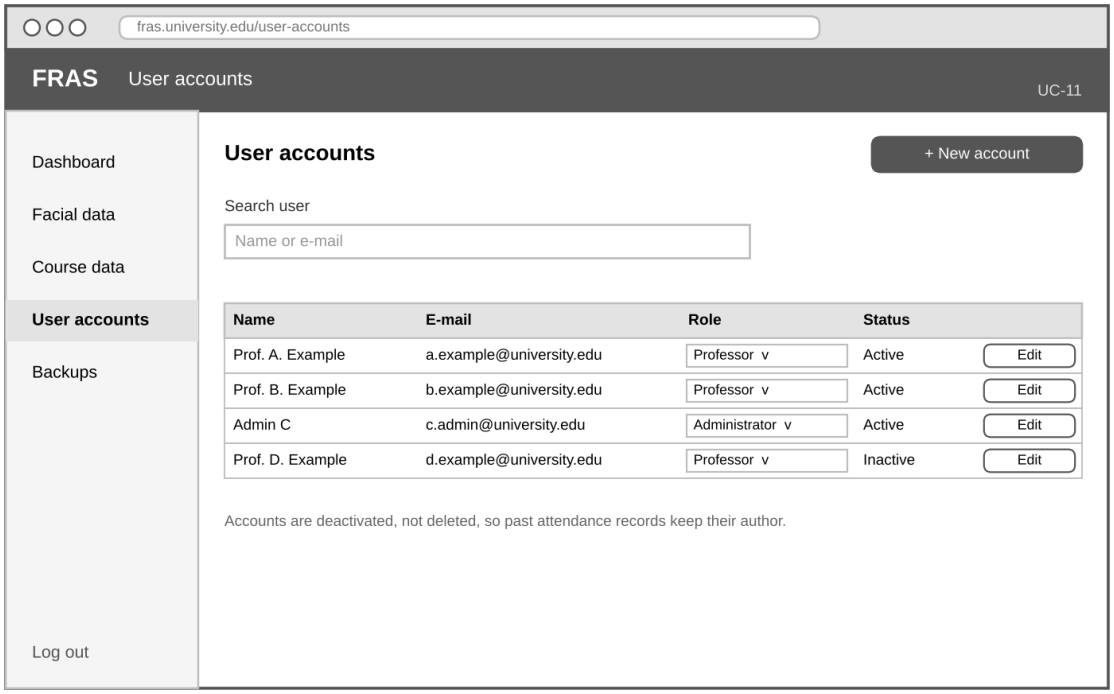


Fig. 12. User accounts (UC-11).

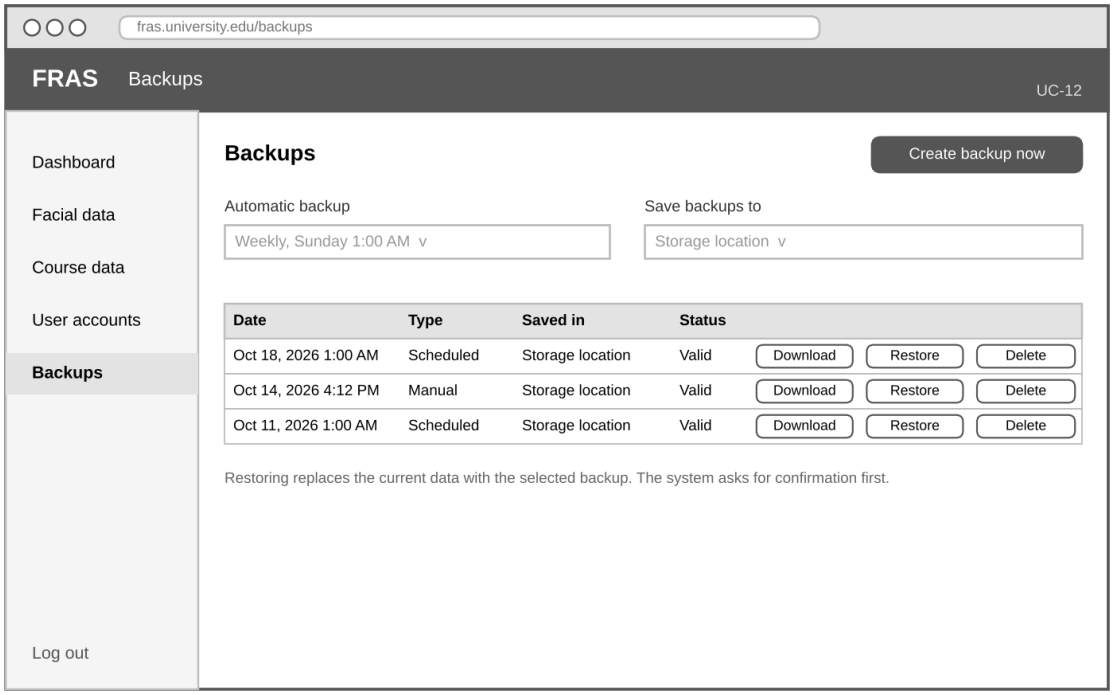


Fig. 13. Backups (UC-12).

TABLE IX
WIREFRAMES AND USE CASES

Screen	Actor	Use cases	Main elements
Log in (Fig. 4)	Both	UC-01	University e-mail, password
My sections (Fig. 5)	Professor	UC-02, UC-07	Sections of the semester; Start session enabled only for the class scheduled now; last session; pending reviews
Open session (Fig. 6)	Professor	UC-02, UC-03, UC-04	Recognized students with time and status; preview of the captured image; summary; Close session
Review (Fig. 7)	Professor	UC-05, UC-06	Editable status and source of each record; flagged faces (Confirm/Reject); manual entry of an authorized student
Reports (Fig. 8)	Professor	UC-07	Section and date filters; attendance percentage; export to Excel or PDF

Screen	Actor	Use cases	Main elements
Dashboard (Fig. 9)	Administrator	UC-08, UC-10, UC-12	Enrollment status; last synchronization; last backup; sessions open now
Facial data (Fig. 10)	Administrator	UC-08, UC-09	Student search in the university roster; consent checkbox; three reference images; delete facial data
Course data (Fig. 11)	Administrator	UC-10	Read-only sections, schedules, professors, and rosters; Sync now; automatic synchronization
User accounts (Fig. 12)	Administrator	UC-11	Create, edit, change role, activate or deactivate accounts
Backups (Fig. 13)	Administrator	UC-12	Create backup now; automatic backup; storage location; download, restore, or delete a backup

VIII. Study of Alternatives

The hardware and software that will implement FRAS have not been selected. Instead of choosing components in advance, the team defined a study of alternatives for the five parts of the system that depend on a technology decision: image capture, the location where images are processed, the face recognition software, data storage, and the connection with the university information system. The options will be compared with the criteria of Table X, which are derived from the requirements and from the evaluation metrics used by Rukhiran *et al.* [5], using a weighted decision matrix. Cost is the main criterion, but no option is accepted if it cannot meet the mandatory requirements for accuracy, response time, and protection of biometric data. Tables XI–XV list the options and the reference data collected so far from manufacturers, service providers, and public benchmarks; these data are a starting point for comparison and will be confirmed before any purchase.

TABLE X
Selection Criteria

Criterion	What is compared	Related requirement
Cost	Purchase cost per classroom and recurring cost (services, licenses, maintenance).	Team constraint (main criterion)
Accuracy	Share of present, enrolled students correctly identified; wrong identifications.	NFR-01 (≥ 90%), NFR-03
Response time	Time from capture to stored record.	NFR-07 (< 3 s)
Privacy and security	Where images and facial data travel and are stored; who can access them.	NFR-46, NFR-50; FERPA, ISO/IEC 24745
Classroom fit	Field of view and resolution to reach the last row; installation; network needs.	NFR-31, NFR-12
Maintainability	Possibility of replacing the recognition model without redesigning the system.	NFR-59
University compatibility	Approval and support from university IT; access to course data.	UC-10, FR-57

TABLE XI
Image Capture Options

Option	Reference data	To study
A1. Camera module for a single-board computer (e.g., Raspberry Pi Camera Module 3 [22])	11.9 MP sensor; US$26.95 (standard lens), US$37.95 (wide lens) [23].	Works only with a compatible board (options B1 and B5); a wide lens may be needed to cover the room.
A2. USB webcam (e.g., Logitech C920s)	1080p at 30 fps; US$59.99 (regular price US$69.99) [24].	Works with any computer; check whether 1080p is enough for faces in the last row.
A3. Network (IP) camera	Sends video over the network to a computer or server.	Price and resolution to be collected; requires a network connection approved by university IT.
A4. Camera module on a one-axis servo (e.g., SG90-type servo controlled by the board)	The servo turns the camera left and right to photograph different parts of the room; proposed by the team to cover more of the room with one camera.	Wider horizontal coverage with one camera; moving parts, strain on the camera cable, no vertical movement, and photos taken at different moments.

TABLE XII
Processing Location Options

Option	Reference data	To study
B1. Single-board computer in the classroom (e.g., Raspberry Pi 5 [25])	US$85 (4 GB) and US$125 (8 GB) in Feb. 2026 [26], with a further increase announced in Apr. 2026 [27]. On a Raspberry Pi 4B: 6.23 ms for face detection (YuNet, 160×120) and 68.82 ms per face for recognition (SFace, 112×112) [28].	Images stay in the classroom; one device per classroom; time for a full class.
B2. AI board in the classroom (e.g., NVIDIA Jetson Orin Nano Super)	US$399 at a U.S. distributor [29]. Jetson Orin Nano: 2.59 ms (YuNet) and 20.05 ms (SFace) on the CPU; 5.23 ms and 7.59 ms on the GPU [28].	Faster than B1 but more expensive per classroom.
B3. Central computer or server on campus	Intel Core i7-12700K CPU: 0.69 ms (YuNet) and 5.09 ms (SFace) [28].	One server for many classrooms; images travel over the campus network; support from university IT.

Option	Reference data	To study
B4. Cloud service (e.g., Amazon Rekognition)	US$0.001 per image for the first million images; 1,000 free images per month during a 12-month free tier; US$0.00001 per stored face vector per month [30]. Face identification in Microsoft Azure requires registration and approval by Microsoft [31].	Images leave the university; depends on the Internet connection; privacy review under FERPA.
B5. Raspberry Pi 3 Model B+ (low-cost board)	Quad-core 64-bit Cortex-A53 at 1.4 GHz [32]; launched at US$35 in 2018 [33]. Not included in the OpenCV Zoo benchmark; slower than the Raspberry Pi 4B.	Lowest cost if the team already has one; recognition of a full class not guaranteed.

TABLE XIII
FACE RECOGNITION SOFTWARE OPTIONS

Option	Reference data	To study
C1. DeepFace	Python framework, MIT license; wraps several recognition models (VGG-Face, FaceNet, ArcFace, Dlib, SFace, GhostFaceNet, among others) and detectors (OpenCV, RetinaFace, MediaPipe, YOLO, YuNet, among others); optional anti-spoofing; can store data in SQLite, PostgreSQL/pgvector, MySQL, and other databases [34].	Proposed by the team as the main base; compare the models it includes on classroom images.
C2. OpenCV with YuNet and SFace	Lightweight detector and recognizer distributed with OpenCV; times in Table XII.	Simplest option for a small device; accuracy in classroom conditions.
C3. face_recognition (dlib)	MIT license; the author reports 99.38% on the Labeled Faces in the Wild benchmark; includes Raspberry Pi examples [35].	Maintenance activity of the project.
C4. Cloud recognition service	See option B4.	Same privacy and cost questions as B4.

TABLE XIV
DATA STORAGE OPTIONS

Option	Reference data	To study
D1. SQLite	File-based database without a server; supported by DeepFace [34].	Suitable for one device per classroom; concurrent access from the web page.
D2. PostgreSQL (with pgvector)	Database server; pgvector stores face embeddings for search; supported by DeepFace.	Suitable for a central server (B3); requires administration.
D3. MySQL	Database server; supported by DeepFace.	Same as D2.

TABLE XV
UNIVERSITY CONNECTION OPTIONS

Option	Reference data	To study
E1. Interface of the student information system (e.g., Ellucian Ethos for Banner or Colleague)	Integration platform for higher education; access through API keys issued for each authorized system [36].	Which system the university uses; whether IT issues keys for a student project.
E2. Interface of the learning management system (e.g., Blackboard Learn REST API)	The endpoint GET /learn/api/public/v1/courses/{courseId}/users returns the members of a course [37].	Whether the system contains schedules and rooms, or only rosters.
E3. Read-only database view provided by university IT	Direct read access to the tables needed by FRAS.	Security policies of the university.

The connection with the university is defined at the design level; its implementation will depend on the access that the university grants. The location of the backups (the same server, an external drive, university storage, or a cloud service) will be decided together with the storage option, because backups also contain facial data. Accuracy figures reported on benchmarks such as Labeled Faces in the Wild are measured on collections of photographs and do not reflect a classroom, where faces are small, at an angle, and under variable light; for that reason, the comparison will give priority to results measured with real users, such as the user tests reported by Ismail *et al.* [8]. The study will follow five steps:

1) Complete the data of every option, including prices on the date of the study.
2) Compare the accuracy and processing time reported for each recognition option in published benchmarks and studies.
3) Check dependencies between options (A1 and A4 require B1 or B5; B4 implies C4) and compare only consistent combinations.
4) Score each option from 1 to 5 on every criterion, with weights agreed with the mentor.
5) Select one option per part and present the decision with its matrix and its data.

IX. NEXT STEPS

The work that follows this report continues the evolutionary prototyping cycle and is ordered so that the system is defined after its users have been heard. Because systems of this kind already exist, what distinguishes FRAS is how well it fits the

way professors work; the requirements are therefore validated with professors before the technology is selected, and the detailed design is made for the selected technology. Table XVI lists the next steps in order.

TABLE XVI
NEXT STEPS

Step	Activity
1	Interviews with professors using the interview guide and the wireframes: session start, make-up classes, number and timing of captures, review of flagged faces, reports, and privacy concerns.
2	Update of the requirements, use cases, and wireframes with the results of the interviews.
3	Study of alternatives: complete the data of each option and apply the criteria of Table X to select the technology for each part of the system.
4	Detailed design of the selected system: flow chart, module diagram, flow graph and cyclomatic complexity [38], entity–relationship model, and class and sequence diagrams.
5	Test design: test cases derived from the independent paths of the flow graph.

X. CONCLUSION

This report presented the progress in the design of FRAS, a system that records class attendance by recognizing the faces of enrolled students. A literature search in IEEE Xplore and Google Scholar located published attendance systems and the technical and regulatory sources that support the design. The requirements specification contains 128 requirements, organized by feature area and by the quality characteristics of ISO/IEC 25010, with measurable targets for accuracy (at least 90%) and response time (less than 3 s). The structural diagram defines seven subsystems independent of any technology; twelve use cases define what the professor and the administrator do with the system and are traced to the functional requirements; and ten wireframes show how each use case is carried out in the web interface. The hardware and software were not chosen in advance: a study of alternatives defines the options for each part of the system and the criteria to select them, with cost as the main criterion. With the interviews and the study of alternatives, the team will validate the design with its users and select the technology on evidence rather than assumption.

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REFERENCES

[1] U.S. Department of Education, "Biometric record," Protecting Student Privacy (FERPA, 34 CFR §99.3). [Online]. Available: https://studentprivacy.ed.gov/content/biometric-record

[2] *Information Security, Cybersecurity and Privacy Protection — Biometric Information Protection*, ISO/IEC 24745:2022, International Organization for Standardization, 2022.

[3] A. Raj, A. Raj, and I. Ahmad, "Smart attendance monitoring system with computer vision using IoT," *J. Mobile Multimedia*, vol. 17, no. 1–3, pp. 115–126, 2021, doi: 10.13052/jmm1550-4646.17135.

[4] Z. Gao *et al.*, "A student attendance management method based on crowdsensing in classroom environment," *IEEE Access*, vol. 9, pp. 31481–31492, 2021, doi: 10.1109/ACCESS.2021.3060256.

[5] M. Rukhiran, S. Wong-In, and P. Netinant, "IoT-based biometric recognition systems in education for identity verification services: Quality assessment approach," *IEEE Access*, vol. 11, pp. 22767–22787, 2023, doi: 10.1109/ACCESS.2023.3253024.

[6] A. Baobaid, M. Meribout, V. K. Tiwari, and J. P. Pena, "Hardware accelerators for real-time face recognition: A survey," *IEEE Access*, vol. 10, pp. 83723–83739, 2022, doi: 10.1109/ACCESS.2022.3194915.

[7] B. Meden *et al.*, "Privacy–enhancing face biometrics: A comprehensive survey," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 4147–4183, 2021, doi: 10.1109/TIFS.2021.3096024.

[8] N. A. Ismail *et al.*, "Web-based university classroom attendance system based on deep learning face recognition," *KSII Trans. Internet Inf. Syst.*, vol. 16, no. 2, pp. 503–523, Feb. 2022, doi: 10.3837/tiis.2022.02.008.

[9] M. Wang and W. Deng, "Deep face recognition: A survey," *Neurocomputing*, vol. 429, pp. 215–244, 2021, doi: 10.1016/j.neucom.2020.10.081.

[10] W. Wu, H. Peng, and S. Yu, "YuNet: A tiny millisecond-level face detector," *Machine Intelligence Research*, vol. 20, no. 5, pp. 656–665, 2023, doi: 10.1007/s11633-023-1423-y.

[11] P. Grother, M. Ngan, and K. Hanaoka, "Face Recognition Vendor Test (FRVT) Part 3: Demographic effects," National Institute of Standards and Technology, Gaithersburg, MD, USA, NISTIR 8280, Dec. 2019, doi: 10.6028/NIST.IR.8280.

[12] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 815–823, doi: 10.1109/CVPR.2015.7298682.

[13] Y. Zhong, W. Deng, J. Hu, D. Zhao, X. Li, and D. Wen, "SFace: Sigmoid-constrained hypersphere loss for robust face recognition," *IEEE Trans. Image Process.*, vol. 30, pp. 2587–2598, 2021, doi: 10.1109/TIP.2020.3048632.

[14] OpenCV, "DNN-based face detection and recognition," OpenCV 4.x Documentation. [Online]. Available: https://docs.opencv.org/4.x/d0/dd4/tutorial_dnn_face.html

[15] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. New York, NY, USA: McGraw-Hill, 2020.

[16] K. C. Laudon and J. P. Laudon, *Management Information Systems: Organization and Technology*, 4th ed. Upper Saddle River, NJ, USA: Prentice Hall, 1996.

[17] *Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model*, ISO/IEC 25010:2023, International Organization for Standardization, 2023.

[18] *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Std 830-1998, 1998.

[19] R. J. Lopez Ortiz and M. A. de Jesús Rosa, "Software requirements specification — Facial Recognition Attendance System (FRAS)." [Online]. Available: https://drive.google.com/file/d/1thL-HD7tyZLLVYdYCgF-m_I5ArkI0Jlk/view?usp=sharing

[20] I. Jacobson, M. Christerson, P. Jonsson, and G. Övergaard, *Object-Oriented Software Engineering: A Use Case Driven Approach*. Wokingham, U.K.: Addison-Wesley, 1992.

[21] M. Cohn, *User Stories Applied: For Agile Software Development*. Boston, MA, USA: Addison-Wesley, 2004.

[22] Raspberry Pi Ltd., "Raspberry Pi Camera Module 3 — Product brief." [Online]. Available: https://www.raspberrypi.com/products/camera-module-3/

[23] CanaKit, "Raspberry Pi Camera Module 3." [Online]. Available: https://www.canakit.com/raspberry-pi-camera-module-3.html.

[24] Staples, "Logitech C920S Pro 1080p HD Webcam (960-001257)." [Online]. Available: https://www.staples.com/product_24383955.

[25] E. Upton, "Introducing: Raspberry Pi 5!," Raspberry Pi Ltd., Sep. 28, 2023. [Online]. Available: https://www.raspberrypi.com/news/introducing-raspberry-pi-5/

[26] PCWorld, "Raspberry Pi prices leap again as RAM shortages slam PCs," Feb. 2, 2026. [Online]. Available:

https://www.pcworld.com/article/3049682/raspberry-pi-prices-leap-again-as-ram-shortages-slam-pcs.html

[27] GIGAZINE, “Raspberry Pi announces further price increases, and a new version of the older model with increased RAM capacity is also being released,” Apr. 2, 2026. [Online]. Available: https://www.gigazine.net/gsc_news/en/20260402-raspberry-pi-price-increases

[28] OpenCV, “OpenCV Zoo and Benchmark,” GitHub repository, benchmark table. [Online]. Available: https://github.com/opencv/opencv_zoo.

[29] SparkFun Electronics, “NVIDIA Jetson Orin Nano Super Developer Kit (DEV-22098).” [Online]. Available: https://www.sparkfun.com/products/22098.

[30] Amazon Web Services, “Amazon Rekognition pricing.” [Online]. Available: https://aws.amazon.com/rekognition/pricing/.

[31] Microsoft, “Limited Access features for Azure AI Face service.” [Online]. Available: https://learn.microsoft.com/legal/cognitive-services/computer-vision/limited-access-identity.

[32] Raspberry Pi Ltd., “Raspberry Pi 3 Model B+.” [Online]. Available: https://www.raspberrypi.com/products/raspberry-pi-3-model-b-plus/

[33] C. Ward, “Raspberry announces the new Raspberry Pi 3 Model B+ with same US$35 price as its predecessor,” Notebookcheck, Mar. 14, 2018. [Online]. Available: https://www.notebookcheck.net/Raspberry-announces-the-new-Raspberry-Pi-3-Model-B-with-same-US-35-price-as-its-predecessor.289234.0.html

[34] S. I. Serengil, “DeepFace: A lightweight face recognition and facial attribute analysis framework for Python,” GitHub repository. [Online]. Available: https://github.com/serengil/deepface.

[35] A. Geitgey, “face_recognition: The world’s simplest facial recognition API for Python,” GitHub repository. [Online]. Available: https://github.com/ageitgey/face_recognition.

[36] Tray.ai, “Ellucian Ethos connector documentation.” [Online]. Available: https://tray.ai/documentation/connectors/service/ellucian-ethos.

[37] Anthology, “Pulling gradebook data and assessment grades,” Blackboard Developer Documentation. [Online]. Available: https://docs.blackboard.com/docs/blackboard/rest-apis/hands-on/pulling-gradebook-data-and-assessment-grades.

[38] T. J. McCabe, “A complexity measure,” *IEEE Trans. Softw. Eng.*, vol. SE-2, no. 4, pp. 308–320, Dec. 1976, doi: 10.1109/TSE.1976.233837.

[39] OpenAI, “ChatGPT,” GPT-5.6. [Online]. Available: https://chatgpt.com/

[40] Anthropic, “Claude.” [Online]. Available: https://claude.ai/